

Anup K. Srivastava, Ph.D.

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PROFESSIONAL APPOINTMENTS

Children's National Research Institute
PhRMA Research Postdoctoral Scientist

Washington DC, USA
Feb 2021- Present

PROFESSIONAL SUMMARY

Translational neurobiologist with deep experience in neurodegeneration biology, human-derived cellular models, and quantitative biomarker assay development. Trained in designing and executing assays across ELISA, MSD, Western, flow cytometry, RT-qPCR, and protein-level readouts. Advanced expertise in alpha-synuclein, amyloid- β , mitochondrial dysfunction, proteostasis failure, inflammatory biomarkers, and pathway-level PD/AD signatures. Strong record of troubleshooting complex assays, integrating statistics, and translating mechanistic insights into clinically relevant biomarkers. Extensive track record in human neuronal cultures, 3D brain organoid-like systems, SH-SY5Y, C6 glioma, primary tissues, and biofluids, supported by high-impact publications. Experienced in assay development for proteasomal degradation markers, post-translational modifications, pathological protein aggregates, mitochondrial readouts, and inflammatory gene signatures.

CORE TECHNICAL EXPERTISE

Immunoassays & Protein Quantification

- ELISA, Western blot, Electrophoresis
- Immunoprecipitation, protein–nanoparticle interaction assays
- Quantification of PTMs (pS129 α -synuclein, ubiquitination, H2AK119ub)

Cellular & Human-Derived Models

- SH-SY5Y neuroblastoma, C6 glioma, iPSC-derived neurons
- 3D neural spheroids / organotypic slice cultures
- Primary neuronal lysates and biofluids
- Mitochondrial membrane potential assays (JC-1)
- Apoptosis/caspase activation imaging
- Oxidative stress, neuroinflammation, mitophagy assays

Genomic & Proteomic Readouts

- RT-qPCR (inflammatory genes, SNCA pathway, PRC2/BMI-1/EZH2 panel)
- Multiplex cytokine profiling
- Proteasomal activity assays
- Amyloid- β aggregation kinetics
- Protein coronas, nanoparticle-interactome characterization

Flow Cytometry

- Multi-color surface/intracellular labeling
- Caspase-3, mitochondrial potential, oxidative stress markers
- Aggregate-associated fluorescence signatures

Translational & Biomarker Experience

- Alpha-synuclein phosphorylation (pS129)
- Neuroinflammation markers (IL-1 β , NF- κ B, IL-6)
- Mitochondrial injury biomarkers ($\Delta\Psi$ m loss, Akt/3PK signaling changes)
- Amyloid- β aggregation, fibrillation, and detection assays
- Quantitative PD & AD models aligned to in vivo pathology
- Assessment of therapeutic modulation in primary tissues & slices

Computational & Statistical Analysis

- MATLAB, Python, GraphPad Prism
- High-content imaging quantification
- PK/PD-aligned sampling, fold-change modelin
- ANOVA, Bonferroni, Tukey's post-hoc, dose–response modeling

EDUCATION

Indian Institute of Science Education and Research (IISER) & Institute of Nano Science and Technology (INST) Mohali, India

- **Ph.D.** in Nanoscience and Nanotechnology (**2014-2020**)
- Dissertation title: “Hybrid Nanostructures-Mediated Biosensing and Epigenetically Controlled Neurotherapeutic Application in Neurodegenerative Disease.”
- Supervisor: Dr. Surajit Karmakar, Professor and Scientist-F at the Institute of Nano Science and Technology, Mohali, India.

Banaras Hindu University, School of Biotechnology, Varanasi, Uttar Pradesh, India

- **M.Sc.** in Biotechnology (**2011-2013**), Marks: 81.29 %.

Deen Dayal Upadhyay Gorakhpur University, Uttar Pradesh, India

- **B.Sc.** Industrial Microbiology, Zoology, and Chemistry (**2006-2009**), Marks: 66.44 %.

PROFESSIONAL EXPERIENCE

PhRMA Research Postdoctoral Scientist

2021- Present

Center for Neuroscience Research, Children's National Research Institute, Washington, DC

- Designed and executed translational assays in human-derived and rodent-derived systems to quantify neurological disease-linked biomarkers.
- Developed ELISA and multiplex immunoassays for protein quantification in neuronal lysates and biofluids.
- Performed 3D neural culture assays, brain-slice biomarker analyses, and multi-parametric fluorescence imaging.
- Analyzed mitochondrial dysfunction, inflammatory pathway activation, and proteostasis failure using RT-qPCR, Western/JESS, and caspase assays.
- Integrated MATLAB/Python workflows to model biomarker trajectories and quantify therapeutic responses.
- Optimized assay sensitivity and reproducibility across multi-site collaborations.
- Mentored trainees in assay design, execution, QC, and data interpretation
- Maintained rigorous ELN documentation and ensured compliance with data integrity standards.

Senior Research Fellow (SRF)

2016- 2020

Institute of Nanoscience and Technology, Mohali, India

- Led mechanistic studies on alpha-synuclein phosphorylation (pS129), ubiquitin-dependent degradation, BMI-1/PRC2 dysregulation, and inflammatory signaling in PD models.
- Developed melatonin/polydopamine nanostructure therapeutics, linking biomarker changes to therapeutic exposure.
- Designed quantitative immunoassays and Western/JESS workflows to measure proteasomal regulation, epigenetic complex activity, and toxic aggregate clearance.
- Used SH-SY5Y and C6 glioma human-derived models, performing multi-color flow cytometry, mitochondrial assays, and protein kinetics profiling.
- Quantified amyloid- β aggregation, nanoparticle-protein interactions, and corona-mediated changes in AD biomarker detection.

Junior Research Fellow (JRF)

2014- 2016

Institute of Nanoscience and Technology, Mohali, India

- Supported molecular biomarker assays, protein expression studies, and gene expression profiling in neurodegeneration-aligned pathways.
- Executed fluorescence-based biosensing, spectral quantification, and molecular kinetics assays.

SCHOLARLY PUBLICATIONS

Research Articles:

1. Atak, Z.; **Srivastava, A.K.**; Gandhi, S.; Park, E.; Williams, J.; Torii, M.; Haydar, T. *Disrupted neurogenesis from basal intermediate precursor cells alters the postnatal neocortex in the TcMAC21 mouse model of Down syndrome.* **Journal of Neuroscience**, 2025, 45 (34). (IF $\approx$ 6.9)
2. **Srivastava, A.K.**; Choudhury, S.R.; Karmakar, S. *Neuronal BMI-1 mediates ubiquitination and proteasomal degradation of alpha-synuclein in Parkinson's disease.* **Neuropharmacology**, 2021, 194, 108372. (IF: 4.7)
3. **Srivastava, A.K.**; Choudhury, S.R.; Karmakar, S. *Near-infrared responsive dopamine/melatonin-derived nanocomposites abrogating in-situ amyloid beta nucleation, propagation, and ameliorating neuronal function.* **ACS Appl. Mater. Interfaces**, 2020, 12(5), 5658–5670. (IF: 9.5)
4. **Srivastava, A.K.**; Dev, A.; Choudhury, S.R.; Karmakar, S. *Curcumin-tethered lanthanum carbonate nanospheres and protein corona enable fluorescence-based detection of amyloid- β aggregates.* **Sensors Actuators B: Chem.**, 2018, 262, 1–9. (IF: 8.4)
5. **Srivastava, A.K.**; Choudhury, S.R.; Karmakar, S. *Melatonin/polydopamine nanostructures for collective neuroprotection in Parkinson's disease therapy.* **Biomaterials Science**, 2019. DOI: 10.1039/C9BM01602C. (IF: 6.6)
6. Singh, P.K.*; **Srivastava, A.K.***; Dev, A.; Kaundal, B.; Choudhury, S.R.; Karmakar, S. *1,3- β -Glucan-anchored, paclitaxel-loaded chitosan nanocarrier enhances anti-glioblastoma stem cell therapy.* **Carbohydrate Polymers**, 2018, 180, 365–375. (Equal first author, IF: 12.5)
7. Yadav, S.K.*; **Srivastava, A.K.***; Dev, A.; Kaundal, B.; Roy Choudhury, S.; Karmakar, S. *Nanomelatonin triggers superior anticancer functionality in malignant glioblastoma cells.* **Nanotechnology**, 2017, 28(36):365102. (Equal first author, IF: 3.5)
8. Sardoiwala, M.N.; **Srivastava, A.K.**; Kaundal, B.; Karmakar, S.; Choudhury, S.R. *Metformin-loaded polydopamine nanoformulation promotes EZH2-mediated proteasomal degradation of phospho- α -synuclein in Parkinson's disease model.* **Nanomedicine**, 2021, 102088. (IF: 5.4)
9. Kaundal, B.; Kushwaha, A.C.; **Srivastava, A.K.**; Karmakar, S.; Choudhury, S.R. *Non-viral nano-delivery targeting EZH2 for precise acute myeloid leukemia therapy.* **J. Mater. Chem. B**, 2020, 8, 8658–8670. (IF: 7.0)
10. Kushwaha, A.C.; Kaundal, B.; Dev, A.; **Srivastava, A.K.**; Soni, J.M.; Karmakar, S.; Choudhury, S.R. *PRT4165 nanocomposite promotes epigenetic retardation through proteasomal depletion of Polycomb in acute myeloid leukemia.* **Applied Materials Today**, 2020, 21, 100847. (IF: 8.3)

11. Kaundal, B.; **Srivastava, A.K.**; Dev, A.; Soni, J.M.; Karmakar, S.; Choudhury, S.R. *Nanoformulation of EPZ011989 attenuates EZH2–c-Myb epigenetic interaction via proteasomal degradation in AML*. **ACS Molecular Pharmaceutics**, 2020. (IF: 4.9)
12. Kaundal, B.; **Srivastava, A.K.**; Sardoiwala, M.N.; Karmakar, S.; Choudhury, S.R. *NIR-responsive indocyanine green–genistein nanoformulation controlling Polycomb machinery for efficient photo-chemo-combotherapy of glioblastoma*. **Nanoscale Advances**, 2019. (IF: 4.7)
13. Dev, A.; **Srivastava, A.K.**; Choudhury, S.R.; Karmakar, S. *Nano-curcumin enhances blue-light photodynamic therapy against glioblastoma stem cells*. **RSC Advances**, 2016, 6, 95165–95168. (IF: 3.8)
14. Sharma, D.; Barhwal, K.K.; Biswal, S.N.; **Srivastava, A.K.**; Bhardwaj, P.; Kumar, A.; Chaurasia, O.P.; Hota, S.K. *Hypoxia-mediated cholesterol oxidation alters BDNF signaling and induces neurodegeneration in the hippocampus*. **J. Neurochemistry**, 2019, 148, 238–251. (IF: 4.7)

Reviews Articles:

1. **Srivastava, A.K.**; Dev, A.; Karmakar, S. *Nanosensors and nanobiosensors in food and agriculture*. **Environmental Chemistry Letters**, 2018, 16(4). DOI: 10.1007/s10311-017-0674-7. (IF: 20.4)
2. Sardoiwala, M.N.; **Srivastava, A.K.**; Karmakar, S.; Choudhury, S.R. *Nanostructures endow neurotherapeutic potential in optogenetics: Current developments and future prospects*. **ACS Chemical Neuroscience**, 2019, 10(8): 3375–3385. (IF: 3.9)
3. Dev, A.; **Srivastava, A.K.**; Karmakar, S. *Nanomaterial toxicity for plants*. **Environmental Chemistry Letters**, 2018. DOI: 10.1007/s10311-017-0667-6. (IF: 20.4)

Book Chapters:

1. **Srivastava, A.K.**; Kaundal, B.; Khanna, G.; Choudhury, S.R.; Karmakar, S. *Covalent and non-covalent functionalization of silica nanoparticles for neurotherapeutic and neurodiagnostic applications*. In: *Functionalized Nanomaterials I*, CRC Press, 2020.
2. **Srivastava, A.K.**; Dev, A.; Karmakar, S. *Nanosensors for food and agriculture*. In: *Nanoscience in Food and Agriculture 5*, Springer, 2017, pp. 41–79. DOI: 10.1007/978-3-319-58496-6_3.
3. Dev, A.; **Srivastava, A.K.**; Karmakar, S. *Uptake and toxicity of nanomaterials in plants*. In: *Nanoscience in Food and Agriculture 5*, Springer, 2017, pp. 169–204. DOI: 10.1007/978-3-319-58496-6_7.

HONORS AND RECOGNITIONS

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| • AES Fellow 2024 Award by American Epilepsy Society | 2024 |
| • International Travel Award by Council of Scientific and Industrial Research (CSIR) for attending “63rd Annual Meeting of the Biophysical Society” organized at Baltimore, USA between 2-6 March 2019. | 2019 |
| • International Travel Award by Department of Biotechnology for attending “63rd Annual Meeting of the Biophysical Society” organized at Baltimore, USA between 2-6 March 2019. | 2019 |
| • Best Research Poster Presentation Award at NANOBIOTECK-2018, organized by Indian Society of Nanomedicine at All India Institute of Medical Sciences (AIIMS) New Delhi, India | 2018 |
| • Junior Research Fellowship (2014), Department of Biotechnology (DBT), Government of India | 2014 |
| • Junior Research Fellowship (JRF), Indian Council of Medical Research (ICMR), Government of India | 2013 |
| • Junior Research Fellowship (JRF), Council of Scientific and Industrial Research (CSIR), Government of India | 2012 |
| • Summer Research Fellowship , Indian Academy of Sciences, India | 2012 |

CONFERENCE PARTICIPATIONS

- Attended the **American Epilepsy Society (AES) Annual Meeting 2024** at Los Angeles, USA, between Dec 5th – 10th 2024.
- **Anup K. Srivastava**, Md. Nadim Sardoiwala, Babita Kaundal, Subhasree R. Choudhury and Surajit Karmakar, “Tailoring Biomolecular Interactions of Hybrid Nanostructure for their Diagnostics and Therapeutic Applications in Neurodegenerative Diseases” at BPS19 a 63rd Annual Meeting of Biophysical Society on Baltimore, MD, the USA between 2nd – 6th March 2019 [podium].
- **Anup K. Srivastava**, Md. Nadim Sardoiwala, Babita Kaundal, Subhasree R. Choudhury and Surajit Karmakar, “Insight into Tethered Nanoparticle-Protein Corona for Sensitive Detection of Amyloid Beta Aggregates” on NANOBIOTECK-2018 a 3rd annual conference of Indian Society of Nanomedicine at AIIMS New-Delhi, India on 25th -7th Oct 2018 [podium].
- **Anup K. Srivastava**, S. K. Yadav, P. Singh, A. Dev, B. Kaundal, S. R. Choudhury and S. Karmakar “Targeting Glioblastoma and Glioblastoma Initiating Stem-like cells with Drug loaded, Surface Functionalized Polymeric Chitosan Nanoparticles” in a CeNS-INST joint symposium at Hospet, Karnataka, India on Oct 2017. [podium and Poster].
- **Anup K. Srivastava**, S. K. Yadav, P. Singh, A. Dev, B. Kaundal, S. R. Choudhury and S. Karmakar “Drug Loaded Polymeric Chitosan Nanoparticles Endows Superior Anti-Glioblastoma Therapeutic Potentials” at 3rd CRIKC Nanoscience Day held on 29 August 2017 at CSIR-CSIO Chandigarh, India [Poster].
- **Anup K. Srivastava**, A. Dev, and S. Karmakar “Curcumin-Icg Loaded Mesoporous Silica Nanovesicles Based Theranostic for Anti-Angiogenic Therapy” at Australia- India Strategic Research Fund Meeting (AISRF2015) between Nov 25th -27th 2015 at INST Mohali, Punjab, India [Poster].
- **Anup K. Srivastava**, A. Dev and S. Karmakar “Amino-Modified Mesoporous Silica Nanovesicles Based Theranostic as Anti-Cancer Therapy” at 6th MRS Trilateral Symposium on Advances in Nanomaterials: Energy, Water & Healthcare” held between 23th to 25th Nov 2015 at INST Mohali, Punjab, India [Poster].
- **Anup K. Srivastava**, D. Narang and S. Mukhopadhyay “Biophysical Studies on a Yeast Prion Determinant: Insights into Strain

Diversity and Species Barrier” Presented poster at “The Second International Symposium on Protein Folding and Dynamics” organized at NCBS, Bangalore, India from 4th -7th Nov 2014 **[Poster]**.

Place: Washington DC

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